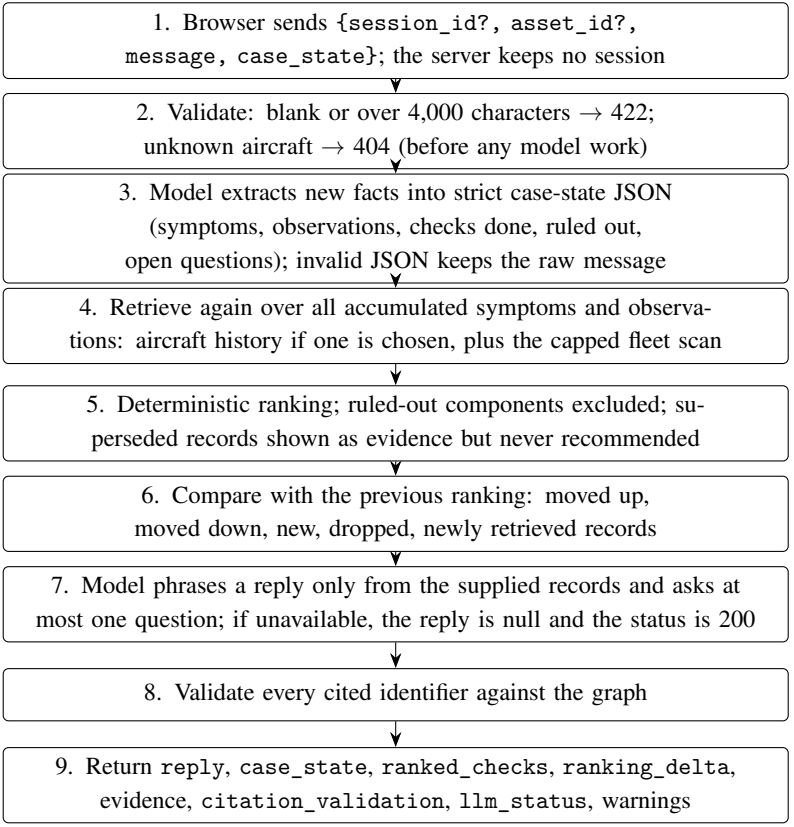
1. Browser sends {session_id?, asset_id?, message, case_state}; the server keeps no session
2. Validate: blank or over 4,000 characters → 422; unknown aircraft → 404 (before any model work)
3. Model extracts new facts into strict case-state JSON (symptoms, observations, checks done, ruled out, open questions); invalid JSON keeps the raw message
4. Retrieve again over all accumulated symptoms and observations: aircraft history if one is chosen, plus the capped fleet scan
5. Deterministic ranking; ruled-out components excluded; superseded records shown as evidence but never recommended
6. Compare with the previous ranking: moved up, moved down, new, dropped, newly retrieved records
7. Model phrases a reply only from the supplied records and asks at most one question; if unavailable, the reply is null and the status is 200
8. Validate every cited identifier against the graph
9. Return reply, case_state, ranked_checks, ranking_delta, evidence, citation_validation, llm_status, warnings